



Survey of the chemical, physical, and sensory characteristics of currently produced mortadella bologna

G. Barbieri^{*}, M. Bergamaschi, Ge. Barbieri, M. Franceschini

Stazione Sperimentale per l'Industria delle Conserve Alimentari, 43100 Parma, Italy

ARTICLE INFO

Article history:

Received 20 July 2012

Received in revised form 16 January 2013

Accepted 7 February 2013

Keywords:

Meat quality

Sensorial analysis

Nutritional aspect

ABSTRACT

In recent years, formulations of the Italian cold cut mortadella have changed to meet evolving tastes and nutritional trends. The aim of the present study was to analyze the physical, chemical, and sensory characteristics of currently produced mortadella. Representative samples of Italian mortadella (13 brands, 3 samples/brand) were analyzed. Three groups of products were identified according to their chemical compositions. The median cluster was composed of 54.80% moisture, 26.19% fat, and 15.02% protein, whereas the other two clusters were characterized by a higher percentage of moisture or fat. These results indicate that, on average, the current products are characterized by less fat (−5%) and salt (−10%) content compared to products 20 years ago.

© 2013 Elsevier Ltd. All rights reserved.

1. Introduction

Mortadella is a common Italian cold cut that traditionally has been made with poor cuts of pork, including both muscle and nonmuscle parts. Most mortadella is produced in North and Central Italy, and the regulation of Mortadella PGI (Protected Geographical Indication), permits production in only these regions. The production of mortadella is characterized by a specific procedure: meat is forced through a special mincing machine, named “exterminator”, obtaining a meat mixture that however is not an emulsion. A particularly long and intense cooking process allows the meat to develop its typical organoleptic characteristics and stabilizes the microbiology of the product, ensuring a long shelf life.

Recently, the formulation of mortadella has changed to meet the evolving tastes of consumers and nutritional trends (Horita, Morgano, Celeghini, & Pollonio, 2011; Trius, Sebranek, Rust, & Carr, 1994; Zanardi, Novelli, Campaninni, Madarena, & Chizzolini, 2000). The use of certain cuts, such as those of the stomach and meat stripped from the head bones, has been dramatically reduced and, in some cases, discontinued entirely. Mortadella producers are becoming increasingly oriented towards products that are lower in fat and salt content.

Mortadella is an economically meaningful product with a large market. In Italy, the annual production of mortadella is as high as 750,000 tons, of which 27,000 tons are exported to the European Community (source: ASSICA, 2011). The dimensions of Mortadella vary significantly, ranging in weight from 1 kg to >50 kg. The most common type of mortadella has an average weight of 12 kg, with a diameter of approximately 24 cm.

The purpose of this survey was to describe the chemical, physical, and sensorial characteristics of the current Italian production of mortadella, in light of recent changes in product formulations. The composition of mortadella was characterized and correlated with sensory and physicochemical properties. This analysis allowed us to evaluate which properties had the greatest influence on quality perception.

2. Materials and methods

To obtain a representative sample, 39 samples of mortadella (of the most common dimension) were obtained from markets located in the north and central regions of Italy. The 13 most well-known brands were chosen, with 3 samples per brand. The goal of the sampling process was to choose typical mortadella products available in Italy. Therefore, mortadella samples were chosen with consideration of the stated composition and price. Discounted products were not considered. To test the widest variety of manufacturing conditions, samples from the same producer (3 samples) came from different lots. The 13 brands were considered to cover more than 80% of the Italian market of mortadella. Each sample underwent the tests described in Sections 2.1 to 2.6.

2.1. Proximate composition and physicochemical analyses

Minced samples were analyzed for moisture, ash, fat, total protein, and collagen (by hydroxyproline content) with the standard AOAC procedure (AOAC, 2000). The approximate composition of the lean mixture was determined after the fat cubes had been removed from the sample. The NaCl concentration was determined from the chloride ion content by the Volhard method (ISO, 1841, 1996). Levels of residual nitrites and nitrates in the products were determined by ion suppression chromatography (Pizza, Pedrielli, Bergamaschi, Barbieri, & Franceschini,

^{*} Corresponding author. Tel.: +39 0521795283.

E-mail address: giampiero.barbieri@ssica.it (G. Barbieri).

2007). The pH of ground samples was measured by insertion electrode (Polilyte Lab) with a pH meter (PU510, XS-Instruments). All analyses were performed in duplicate.

2.2. Total and nitric oxide-haem pigment measurements

The samples were minced after the fat cubes had been removed. Hornsey's method (Hornsey, 1956) was used to determine the concentrations of total pigments (in milligrams of acid haematin per kilogram of mixture) and nitric oxide (NO) pigments (in milligrams of NO-haem per kilogram of mixture) in each sample.

2.3. Colour test

Colour was measured on the lean component of a 1-cm-thick slice of mortadella by spectrocolorimetry (Minolta CM2002 spectrocolorimeter). The operating conditions were as follows: illuminant D₆₅, measurement area of 11 mm, observation angle of 10°, and specular component excluded. The colour coordinates were calculated in the CIE Lab system (CIE, 1976), and the results were expressed as lightness (L*), redness (a*), and yellowness (b*).

2.4. Fatty acid profile

Fat was extracted from a minced sample of mortadella by the Folch method (1956), and 40 mg of fat was methylated with a 2% solution of HCl in methanol as previously described (attachment Xb to CEE norm 2568/91).

Methyl-esters were separated with a gas chromatography system (Trace Ultra, Thermo Fisher), which was fitted with a split/splitless injector and FID detector, on which a SP2380 column (Supelco) had been mounted. The operating conditions were as follows: split injection mode (splitting ratio 75), injected volume of 1 µl, injector and detector temperatures of 240 °C, and carrier flow of 1.3 ml/min. The following procedure was followed: after an initial temperature of 100 °C for 2 min, the temperature was increased to 190 °C at 2 °C/min and maintained at 190 °C for 6 min. The temperature was increased to 220 °C at 15 °C/min and maintained at 220 °C for 15 min. Single methyl-esters were identified by comparison with an external standard (FAME Mix C4–C24, Supelco). Peak integration was performed with ChromQuest version 4.2. Quantities of the single methyl-esters were expressed as the percentages (%) of the total area.

2.5. Texture measurements

Texture was measured with an Instron Universal Testing Machine (mod. 5564) equipped with a 100-kN load cell and interfaced with Bluehill software for data acquisition and processing. For all textural parameters, the mean of 4 measurements per sample was obtained. The compression test, texture profile analysis (TPA), and Warner–Bratzler shear test were carried out at room temperature (25 °C). Specimens (1 cm³) for these 3 textural tests were taken from the lean part of 1-cm-thick mortadella slices by using a hollow metallic sampler with a cubic cutting profile (length = 1 cm).

The specimen was compressed to up to 80% of its original height at a head speed of 50 mm/min. Young's modulus (MY10; in kPa), which reflects the rigidity of an elastic material, was measured as the stress/deformation ratio in the 10% deformation interval, in which the sample can be considered as elastic. The fracturability was determined as the % deformation, which corresponded to the breaking of the specimen along a plane positioned at 45° with respect to the force application axis.

TPA was performed with 40% of the original height of the specimen in two consecutive compression cycles (crosshead speed of 50 mm/min). From the force–time deformation curve, the following parameters were determined (Bourne, 1978): hardness (N), defined as the maximum

force required to compress the sample in the first cycle; and springness (mm), defined as the distance of sample recovery after the first compression.

The Warner–Bratzler shear test was performed with the method suggested by Honikel (1997). Cubic specimens (length = 1 cm) were sheared with a 3-mm-thick steel blade that had a 73° V cut into its lower edge (e.g., like a guillotine with a “V” cut into the blade). The blade travelled through the samples at 100 mm/min. From the force–deformation curve, the maximum load (N) was recorded, which corresponded to the cut of the specimen (shear force).

2.6. Sensorial evaluations

In a preliminary session, the panelists were standardized on their assessment of the following attributes:

1. Consistency (Rc): Resistance to compression of the whole product; range from 0 (very tender) to 9 (very hard)
2. External colour (Ce): Range from 0 (unacceptable or anomalous spots) to 9 (very pleasant, brick red)
3. Sliceability (Sl): Ability of the slice to maintain its shape during slicing operations; range from 0 (slice crumbles) to 9 (good integrity of slice)
4. Internal colour (Ci): Range from 0 (unpleasant, pale, or anomalous) to 9 (very pleasant, bright pink)
5. Internal gelatin pockets (Gp): Separation of gelatin from meat mixture; range from 0 (many pockets with gelatin) to 9 (no gelatin)
6. Internal fat pockets (Fp): Separation of fat from meat mixture; range from 0 (many pockets with fat) to 9 (no separation of fat)
7. Greasiness of slices (Gr): Presence of fat layer on the product after slicing; range from 0 (evidence of fat layer on the surface of the slice) to 9 (no evidence of a fat layer on the surface)
8. Adhesion of fat cubes (Ad): Range from 0 (all of the fat cubes were removed during product slicing) to 9 (all of the fat cubes adhered to the lean portion)
9. Resistance to chewing (Tx): Texture of the product to tooth pressure; range from 0 (very soft during chewing) to 9 (hard during chewing).

The scale is progressive hedonic: a score of 9 always indicates the best sensory appearance, and 0 always indicates the worst one. A trained, 8-member panel evaluated, in a “blind” tasting, three randomized products in each session. Sensory assessments were performed with the panelists separated from each other. Three slices, taken from the central part of the product, were given to each panelist. The mean of eight values were reported.

2.7. Statistics

Data from the chemical, physical, and sensorial analyses were processed with the statistical software package SPSS 14.0. All analyses were performed in duplicate, with the exception of the textural analysis, which was repeated 4 times due to its variability.

The Pearson's correlation among chemical composition, instrumental parameters and sensory evaluation was performed.

In addition to basic statistical analyses, the composition (i.e., moisture, fat, and protein components) of the product was subjected to k-means cluster analysis with subsequent one-way between-groups ANOVA using the last difference square method (LSD).

3. Results and discussion

Mortadella is a mixed-meat product whose characteristics are largely independent of the specific properties of the raw materials, breed of animal, and breeding method. Instead, its characteristics are significantly affected by the formulation and amounts of the cuts used to make it, as well as some technological aspects, such as the use of certain

Table 1

Chemical composition of all samples collected. (Data refer to the whole product and lean mixture).

Sampling	Statistics	Moisture %	Protein %	Fat %	Ash %	NaCl %	pH	Nitrites mg/kg	Nitrates mg/kg	Collagen %
Whole product	Average	55.44	14.60	25.79	3.02	2.26	6.22	5	40	2.17
	S.D.	3.10	1.04	3.23	0.22	0.21	0.12	4	12	0.64
	Minimum	49.96	12.25	19.17	2.61	1.87	5.93	0	23	0.77
	Maximum	61.90	16.85	31.94	3.35	2.81	6.41	16	80	3.20
Lean mixture	Average	61.39	16.20	18.21	3.33	n.d.	n.d.	n.d.	n.d.	2.17
	S.D.	3.02	1.08	2.91	0.30	n.d.	n.d.	n.d.	n.d.	0.55
	Minimum	54.65	13.69	12.44	2.77	n.d.	n.d.	n.d.	n.d.	1.19
	Maximum	67.68	18.26	24.99	3.77	n.d.	n.d.	n.d.	n.d.	3.20

ingredients and additives, and the processing parameters (temperatures and process timing).

Table 1 shows the proximate compositions, pH values, and the nitrite, nitrate, collagen, and sodium chloride contents for the samples. The variability of ash and proteins was relatively low. Fat and moisture comprised most of the composition of the mortadella samples. Products with high moisture contained little fat and vice versa (Pearson's coefficient between fat and moisture of 0.934, $p < 0.005$). Therefore, the different types of mortadella showed varying compositions, between products that were high in moisture and products that were high in fat. The residue of nitrites was always very low, whereas the content of nitrates was quite high (> 60 mg/kg) in three samples only.

Table 2 shows the most significant correlations between some chemical parameters and the sensory parameters. Sensory evaluations referring to the compactness of the mortadella structure, such as its resistance to chewing (Tx) and cohesiveness (Rc), were not correlated in a simple way to the moisture content ($R^2 = 0.191$ and 0.008 , respectively). These sensory characteristics probably depend in a complex way on many factors, including the protein content.

The protein content, which varied in a limited range (4%), was very important in the qualitative product description. All of the sensory variables examined were significantly correlated with this parameter. The protein and collagen contents were both well-correlated with Rc. Moisture and fat, if they were included in a sufficient protein matrix, did not have a negative effect on texture. Conversely, the fat/protein ratio was inversely proportional to the cutting strength (Fig. 1), because the lipidic phase, which might make the product significantly softer, is able to be stabilized if there is a sufficient protein content.

The correlation coefficients in Table 3 indicate that the final composition of mortadella affected the textural characteristics (Cavestany, Jimenez, & Solas, 1994). Correlations were obtained with the compositional data of the lean mixture, from which the instrumental measurements for the sample were taken. Italian mortadella is not an emulsion but is a fine-grain product with plastic rather than elastic compactness features. This characteristic was reflected by the significant correlations that were found between the composition and Young's modulus (MY10). MY10 measures the strength of the sample in the 0–10% interval of specimen deformation. The MY10 was significantly related to almost all of the compositional variables, whereas the instrumental elasticity was negatively related to the collagen content only. Almost all of the components contributed to the product plasticity, which was reduced by a high moisture composition. The positive contribution of protein was mainly due to muscle proteins ($p < 0.001$). This effect is evidenced by the

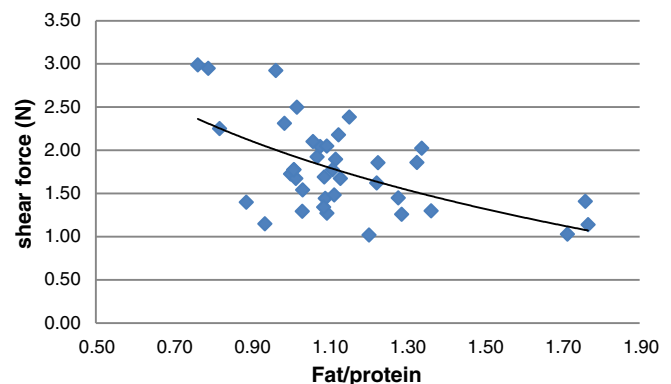


Fig. 1. Effect of the fat/protein ratio on the shear force measured on a sample cube of mortadella (1 cm side).

separation of protein components into their collagen and collagen-free muscle protein parts.

After thermal treatment, the product network was described by its hardness and shear force. In the lean cuts used in product formulation, hardness ($p < 0.01$) and resistance to cutting ($p < 0.05$) were determined from the composition of protein (especially myofibrillar protein). The hardness was reduced by a high moisture/protein ratio ($p < 0.01$), whereas the resistance to cutting was reduced by increased amounts of fat or an elevated fat/protein ratio in the lean part of the product. The salt-soluble protein content was the major factor affecting the amount of fat that could be stabilized. Fat and water are physically or chemically trapped inside the protein matrix (Smith, 1988). Soluble muscle protein encapsulates insoluble lipid and other components to form a coherent matrix.

The percentage of fat also affected the sensory characteristics associated with product texture (Table 4). For example, the consistency and resistance to chewing were higher in samples that were richer in saturated fatty acids (SFAs), whereas they were lower in samples with a higher quantity of unsaturated fatty acids. The fatty acids that had the strongest influence on these sensory parameters were palmitic acid, stearic acid, and linoleic acid. Monounsaturated fatty acids had no particular influence on the sensory parameters.

Fat characterized by a higher percentage of polyunsaturated fatty acids (PUFAs) tends to be softer and is more easily released by the protein network where it is entrapped. This rationale explains the highly

Table 2

Pearson's correlation between chemical composition and sensory parameters.

	Rc	Ce	Gp	Fp	Ci	Ad	Sl	Gri	Tx
Moisture	0.008	0.060	−0.086	0.117	0.357*	0.012	0.104	0.431**	0.191
Fat	−0.097	−0.131	−0.105	−0.314	−0.390*	−0.173	−0.241	−0.577**	−0.333*
Protein	0.506**	0.434**	0.374*	0.456**	0.414**	0.397*	−0.475**	0.617**	0.628**
Collagen	0.318*	0.534**	−0.083	−0.025	0.480**	0.002	0.403*	0.315	0.354*

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 3

Pearson's correlation between instrumental textural parameters and compositional data of a lean mixture of mortadella and sensory evaluations.

	Shear force (N)	MY10 kPa	Hardness (N)	Springness (mm)
Moisture (g/100 g)	0.40*	−0.45**	0.15	0.06
Fat (g/100 g)	−0.51***	0.34*	−0.34*	0.07
Proteins (g/100 g)	0.29	0.50**	0.61***	−0.18
Collagen (g/100 g)	−0.12	0.11	0.14	−0.37*
CFMP (g/100 g)	0.40*	0.46**	0.61***	0.04
Moisture/proteins	0.01	−0.57***	−0.40**	0.18
Fat/proteins	−0.52***	0.13	−0.47**	0.12
RC	0.45**	0.16	0.46**	0.14
SL	0.38*	0.26	0.51**	−0.11
TX	0.53***	0.25	0.67***	0.03

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

negative correlations of the percentage of PUFAs (especially linoleic acid) with the presence of fat pockets in the meat mixture, the adhesion of fat cubes, and the absence of a greasy coat on the slice. The greasy coat on the slice was positively affected by the percentage of SFAs (especially of palmitic and stearic acids), whereas the other two sensory parameters (i.e., fat pockets and adhesion of fat cubes) did not depend on this amount. In particular, the ratio of C18:0 to C18:2 was well-correlated with all of the structural characteristics of mortadella.

Cluster analysis was performed to define the homogeneous sample groups in terms of their compositional characteristics (i.e., moisture, protein, and fat). Three clusters were identified, each with different average values with respect to some of the variables (Table 5). The average values of the sensory values are also shown. The first, most numerous cluster included samples that were high in protein and collagen contents. The second cluster was composed of products that were leaner but higher in moisture. Cluster 3 included samples that were high in fat, low in protein, and had a less-bright colour, characteristics that can be associated with low-quality groups. All of the parameters of mortadella in cluster 3 were different from those of products in clusters 1 and 2 ($p < 0.05$), with the exception of the L^* and b^* values.

The sensory evaluation of colour was confirmed by instrumental data. The a^* value accounted for differences between cluster 3 and all other products. The products in cluster 3 were penalized in terms of Rc and Tx which define the texture. The excess fat content of cluster 3 was highlighted by the Gr and Fp values, although the difference of Fp in cluster 3 compared to the other clusters did not reach statistical significance.

Samples from cluster 3 were differentiated from the other 2 groups in terms of the quality of the total fat (Table 6). In cluster 3, a higher percentage of PUFAs and lower percentage of SFAs penalized these samples in terms of their sensory parameters, such as the presence of fat pockets in the meat mixture, the adhesion of fat cubes, and the fat coat on the slice. The percentage increase of fat in the different samples of mortadella corresponded to an increase in the percentage of PUFAs; products in cluster 3 were high in fat and, therefore, had a significantly higher percentage of PUFAs.

Table 4

Pearson's correlations between fatty acid percentages and sensorial parameters (excluded colour evaluations).

	C16:0	C18:0	C18:2	C16:1	Σ SFA	Σ MUFA	Σ PUFA	P/S	C18:0/C18:2
Rc	0.71***	0.53***	−0.59***	0.51***	0.63***	−0.068	−0.58***	−0.49***	0.65***
Fp	0.23	0.26	−0.68***	0.29	0.24	0.12	−0.65***	−0.59***	0.56***
Ad	0.25	0.20	−0.75***	0.42**	0.23	0.28	−0.72***	−0.69***	0.58***
Sl	0.67***	0.59***	−0.65***	0.34*	0.63***	0.02	−0.64***	−0.56***	0.70***
Gri	0.45***	0.49***	−0.60***	0.34*	0.47***	−0.14	−0.58***	−0.46***	0.63***
Tx	0.62***	0.58***	−0.61***	0.39**	0.59***	−0.03	−0.61***	−0.52***	0.69***

*** $P < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 5

Chemical analysis, instrumental colour, and sensorial data of mortadella sausages for each group identified by cluster analysis (least squares difference test).

	Cluster 1		Cluster 2		Cluster 3		p
	Average	S.D.	Average	S.D.	Average	S.D.	
Moisture %	54.80 ^b	1.73	60.13 ^a	1.39	51.85 ^c	1.86	***
Fat %	26.19 ^b	1.81	21.21 ^a	1.38	30.26 ^c	1.53	***
Protein %	15.02 ^a	0.85	14.26 ^b	0.76	13.31 ^c	0.90	**
Collagen %	2.47 ^a	0.47	1.75 ^b	0.71	1.48 ^b	0.22	***
Salt %	2.33 ^a	0.21	2.07 ^b	0.11	2.22 ^a	0.14	***
L^*	63.55 ^a	1.67	64.16 ^a	0.88	64.05 ^a	3.12	n.s.
a^*	9.08 ^a	0.91	9.09 ^a	0.94	8.13 ^b	1.54	*
b^*	12.59 ^a	1.18	12.37 ^a	0.87	13.56 ^a	0.56	n.s.
Rc	8.45 ^a	0.76	8.03 ^a	0.83	6.75 ^b	0.27	***
Ce	7.83 ^a	0.55	7.37 ^a	0.86	6.35 ^b	0.48	***
Sl	9.53 ^a	0.45	9.36 ^a	0.84	7.87 ^b	1.18	***
Gp	9.49 ^a	0.56	9.48 ^a	0.22	8.87 ^b	0.82	*
Ci	7.72 ^a	0.41	7.70 ^a	0.27	6.50 ^b	0.28	***
Gri	8.59 ^a	0.54	8.67 ^a	0.32	5.67 ^b	0.46	***
TX	8.34 ^a	0.76	7.98 ^a	0.83	6.10 ^b	0.51	***
Fp	9.30 ^a	0.59	9.44 ^a	0.37	8.15 ^b	0.82	***
Ad	8.70 ^a	0.84	8.7 ^a	0.56	7.52 ^b	0.86	***
No of samples	25		8		6		

*** $P < 0.001$; ** $p < 0.01$; * $p < 0.05$; a,b,c: different letters in the same row correspond to significant difference among clusters, n.s.: not significant.

In terms of texture, the products from cluster 3 were characterized by parameters of hardness and shear force that were considerably different from those of the two other clusters. This higher fat composition was reflected in a softer product with a lower resistance to cutting. The sensory evaluations of clusters 1 and 2 did not differ, whereas mortadella in cluster 3 was heavily penalized in the judgments of the panelists.

Table 7 compares the compositions found in the present study with those from an analogous study done in 1990, which considered 17 brands and 34 different products (Pedrielli, Pizza, personal communications). The most diffuse types of Italian mortadella were

Table 6

Fatty acid composition and texture measurements of mortadella sausages for each group identified by cluster analysis. (Least square difference test).

	Cluster 1		Cluster 2		Cluster 3		p
	Average	St. d.	Average	St. d.	Average	St. d.	
C18:2 %	12.43 ^a	0.76	12.46 ^a	0.67	13.47 ^b	0.55	**
Σ SFA %	35.91 ^b	0.96	36.74 ^c	0.79	34.95 ^a	0.4	***
Σ MUFA %	48.3	0.61	48.16	0.7	48.83	0.44	n.s.
Σ PUFA %	14.37 ^a	0.87	14.35 ^a	0.77	15.43 ^b	0.58	**
P/S	0.32	0.02	0.32	0.02	0.34	0.02	n.s.
C18:0/C18:2	0.91 ^b	0.09	0.93 ^b	0.06	0.79 ^a	0.04	***
Shear Force (N)	1.77 ^b	0.40	1.89 ^b	0.54	1.26 ^a	0.17	**
MY10 (kPa)	2094 ^b	663	1552 ^a	156	2137 ^{ab}	323	*
Hardness (N)	14.84 ^b	2.69	13.37 ^b	2.18	11.20 ^a	0.39	**
Springiness (mm)	3.53 ^a	0.70	3.74 ^a	0.74	3.59 ^a	0.13	n.s.

*** $P < 0.001$; ** $p < 0.01$; * $p < 0.05$; a,b,c: different letters in the same row correspond to significant difference among clusters.

Table 7

Comparison between the compositions of mortadella samples analyzed in two different years: 1990 (Pedrielli, Pizza, personal communications) and 2010 (the present study).

		1990	2010	p
Moisture %	Average	54.44	55.44	n.s.
	S.D.	1.72	3.10	
	Min.	51.27	49.96	
	Max	57.93	61.90	
Ash %	Average	3.47	3.02	***
	S.D.	0.22	0.23	
	Min.	3.12	2.61	
	Max	3.81	3.35	
Fat %	Average	27.21	25.79	*
	S.D.	1.88	3.23	
	Min.	23.16	19.17	
	Max	30.24	31.94	
Proteins %	Average	13.83	14.60	**
	S.D.	0.63	1.04	
	Min.	12.87	12.25	
	Max	14.87	16.85	
Collagen %	Average	1.93	2.17	n.s.
	S.D.	0.58	0.65	
	Min.	1.06	0.77	
	Max	3.14	3.20	
Salt %	Average	2.51	2.26	***
	S.D.	0.18	0.21	
	Min.	2.34	1.87	
	Max	2.87	2.81	
Pigments mg/kg	Average	72.49	70.77	n.s.
	S.D.	5.63	8.87	
	Min.	63.28	53.72	
	Max	83.01	89.76	

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

characterized on the basis of their chemical compositions. The salt and fat contents of mortadella in our study were decreased dramatically (by 10% and 5%, respectively) compared to the results from 1990. This decrease was compensated by an increase in protein (+5.5%) and moisture contents. In general, the comparison revealed that current formulations are oriented towards an increased use of muscle cuts with fewer fat cuts and less use of meat stripped from the head bones or muscles.

A similar comparative study was done in 1971, which compared analytical data on mortadella produced in 1948, 1965, and 1970 (Cavallaro, Apa, & Allavena, 1971). That study revealed that the quality of mortadella had significantly worsened, due to the inclusion of a higher percentage of collagen in the sausage. Nowadays the current trend seems to be the production of lighter mortadella.

4. Conclusion

Samples of the analyzed products were grouped into 3 clusters on the basis of their chemical compositions. The sensory parameters, which most correspond to a qualitative classification on analytical

basis, included resistance of the entire piece (of sausage) to compression, resistance of one slice to chewing, greasiness perceived at tasting, and colour. The screening of mortadella sausages that are currently produced in Italy revealed a significant change in the composition over the last 20 years. The salt and fat contents have decreased substantially to meet the current needs of consumers and recent recommendations of nutritional and health experts. This change has led to new formulations of this cold cut, such that producers today prefer to use protein-rich cuts and to avoid fatty cuts. The reduction of fats, especially SFAs, from mortadella underlies the technological and sensory problems that have been emphasized in previous studies (Bergamaschi et al., 2003; Pizza et al., 2004). Nevertheless, it is possible to obtain a very low fat content (approximately 12% in lean mixture and 19% where fat cubes are present) without using replacements that negatively affect the sensory characteristics of mortadella (Zanardi et al., 2000).

References

- AOAC (2000). *Official Methods of Analysis of AOAC International* (17th ed.). Arlington, Virginia, USA: Association of Official Analytical Chemistry.
- ASSICA (2011). *Annual Report*.
- Bergamaschi, M., Pizza, A., Pedrielli, R., Santi, S., Franceschini, M., & Gianni, C. (2003). Production technology of PGI Mortadella Bologna sausages: effect of swine diet and storage conditions on the quality of frozen material and sausages. *Industria Conserve*, 78(3), 251–265.
- Bourne, M. C. (1978). Texture profile analysis. *Food Technology*, 32, 62–66.
- Cavallaro, A., Apa, G., & Allavena, A. (1971). *Bollettino dei Laboratori Chimici Provinciali*, 22(4), 687–704.
- Cavestany, M., Jimenez, Colmenero F., & Solas, M. T. (1994). Incorporation of sardine surimi in Bologna sausage containing different fat levels. *Meat Science*, 38, 27–37.
- CIE (1976). *Recommendation on uniform color spaces-color difference equations*. Psychometric Color Terms. Commission internationale de l'Eclairage. Paris, France. Supplement No 2 to CIE Publication No 15 (E-1.3.1.).
- Honikel, K. O. (1997). Reference methods supported by OECD and their use in Mediterranean meat products. *Food Chemistry*, 59, 573–582.
- Horita, C. N., Morgano, M. A., Celeghini, R. M. S., & Pollonio, M. A. R. (2011). Physico-Chemical and sensory properties of reduced-fat mortadella prepared with blends of calcium, magnesium and potassium chloride as partial substitutes for sodium chloride. *Meat Science*, 89, 426–433.
- Hornsey, H. C. (1956). The colour of cooked cured pork. I. Estimation of nitric oxide-haem pigments. *Journal of the Science of Food and Agriculture*, 7, 534–540.
- ISO 1841 (1996). *Meat and meat products – Determination of chloride content – Part 1: Volhard method*.
- Pizza, A., Pedrielli, R., Barbieri, G., Bergamaschi, M., Gianni, C., & Franceschini, M. (2004). Adding value to the formulations of typical PGI products (mortadella Bologna and zampone Modena): effects of raw materials and storage conditions on the formation of the quality characteristics of the products. *Industria Conserve*, 79(4), 405–423.
- Pizza, A., Pedrielli, R., Bergamaschi, M., Barbieri, G., & Franceschini, M. (2007). Residual nitrite and nitrate in preserved meat products. *Industria Conserve*, 85(4), 405–423.
- Smith, D. M. (1988). Meat proteins: functional properties in comminuted meat products. *Food Technology*, 116–121.
- Trius, A., Sebranek, J. G., Rust, R. E., & Carr, J. M. (1994). Low-Fat Bologna and Beaker sausage: effects of carrageenans and chloride salts. *Journal of Food Science*, 5, 941–945.
- Zanardi, E., Novelli, E., Campaninni, G., Madarena, G., & Chizzolini, R. (2000). *Low-fat mortadella: experimental formulations with some fat substitutes*, 19. (pp. 317–326) Parma: Annali della Facoltà di Medicina Veterinaria.